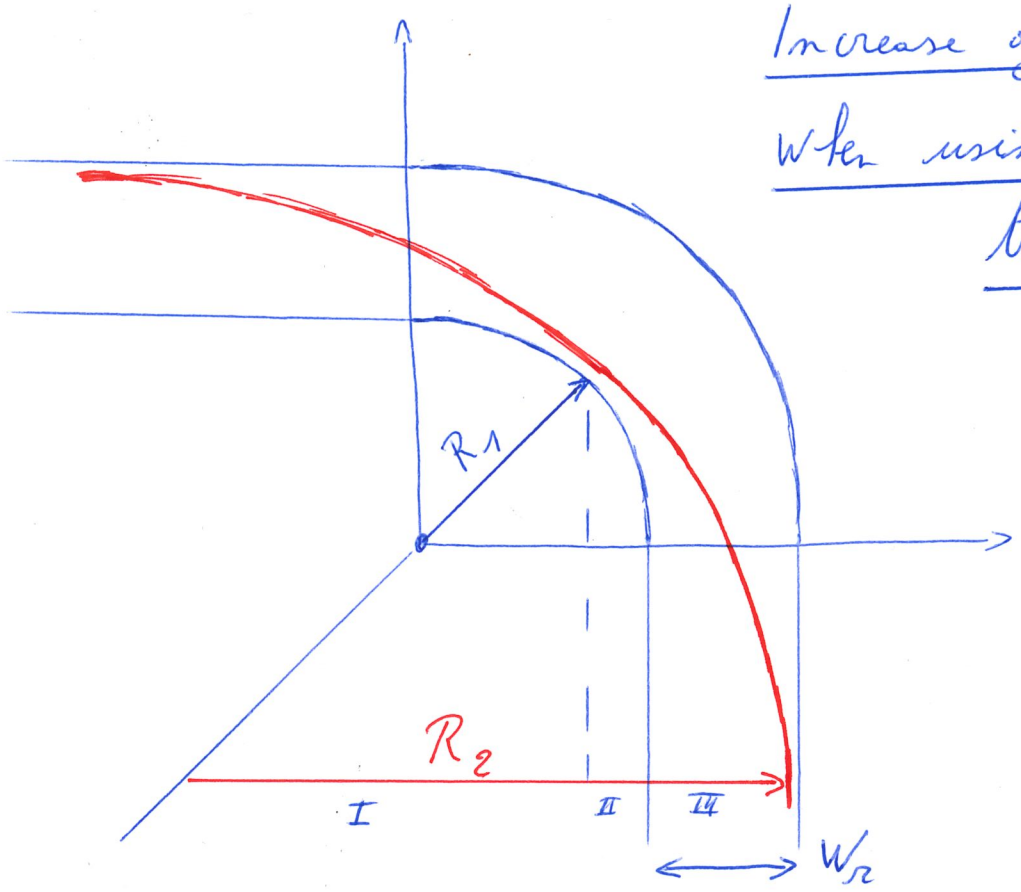
Increase of the speed when using the best trajectory



$$R_2 = R_2 \cos 45 + R_1 (1 - \cos 45) + W_2$$

$$(R_2 - R_1)(1 - \cos 45) = W_2$$

$$R_2 - R_1 = \frac{W_2}{1 - \cos 45} \Rightarrow \boxed{R_2 = R_1 + \frac{W_2}{0{,}2929}}$$

Or $\gamma = V^2/R \Rightarrow V = \sqrt{\gamma R}$

$\gamma = 0{,}7\ g$
$R_1 = 30\ m$
$W_2 = 6\ m$

$R_2 = 50\ m$

$\Rightarrow V_1 = \sqrt{0{,}7g \cdot 30} \times 3{,}6 \simeq 52$ kph.

$V_2 = \sqrt{0{,}7g \cdot 50} \times 3{,}6 \simeq 67$ kph